

DATA VISUALIZATION78

```
!pip install matplotlib seaborn pandas numpy
```

```
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (2.0.2)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cyclers>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.1)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
```

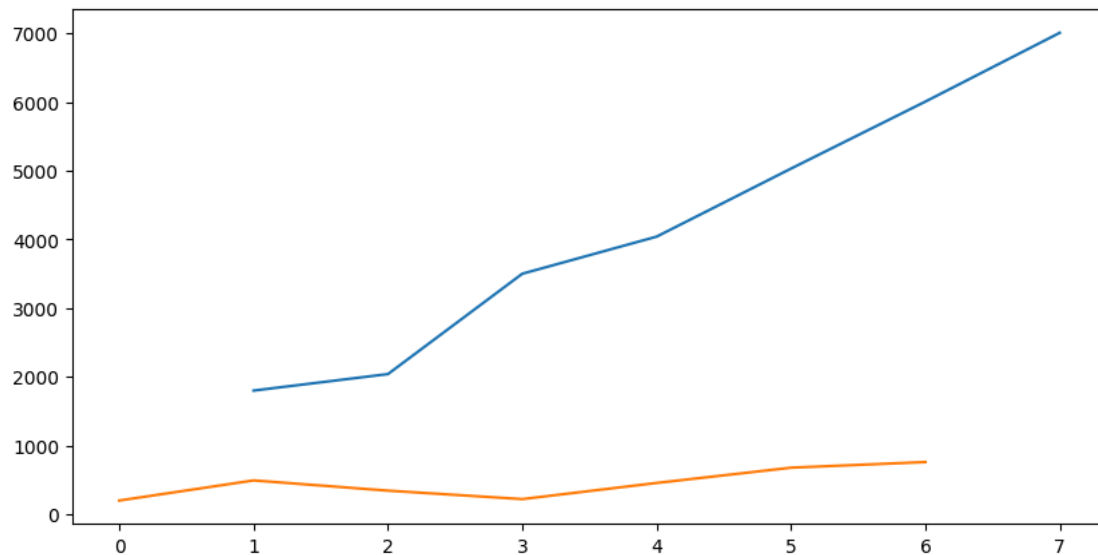
```
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np
```

```
age=[1,2,3,4,5,6,7]
salary=[1800,2040,3500,4040,5030,6005,7006]
id=[200,493,344,221,456,678,760]
plt.figure(figsize=(10,5))
plt.plot(age,salary,id)
plt.plot(age,salary,id,marker="+",label="salary",color="blue",linewidth=1.5)
plt.title("age vs salary")
plt.xlabel("age")
plt.ylabel("salary")
plt.grid(True, alpha=0.3)
plt.savefig("age_vry.png")
plt.show()
```



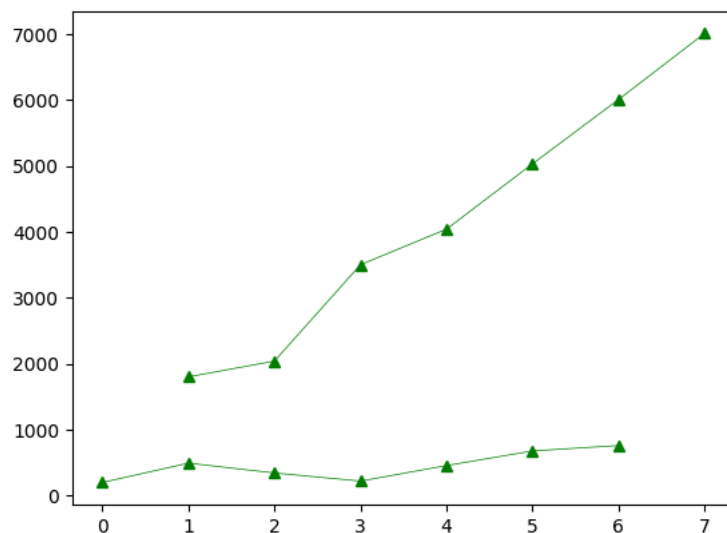
```
plt.figure(figsize=(10,5))
plt.plot(age,salary,id)
```

```
[<matplotlib.lines.Line2D at 0x7925ca98d370>,  
<matplotlib.lines.Line2D at 0x7925ccccf9c40>]
```



```
plt.plot(age,salary,id,marker="^",label="age",color="green",linewidth=0.5)
```

```
[<matplotlib.lines.Line2D at 0x7925ca40cd70>,  
<matplotlib.lines.Line2D at 0x7925ca40d040>]
```



```
import matplotlib.pyplot as plt
```

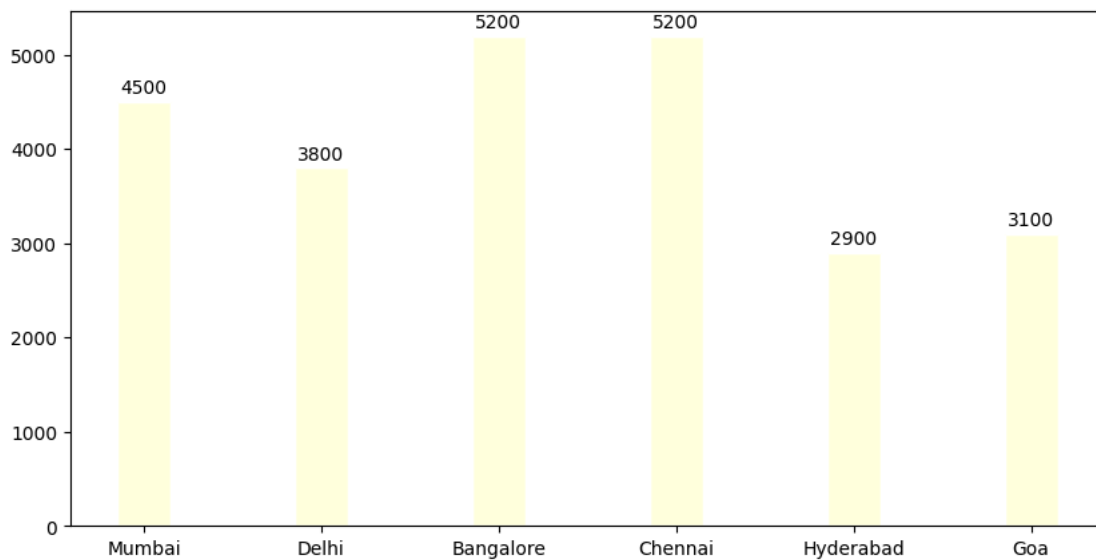
```
cities = ['Mumbai','Delhi','Bangalore','Chennai','Hyderabad','Goa']  
revenue = [4500,3800,5200,5200,2900,3100]
```

```
fig, ax = plt.subplots(figsize=(10,5))
```

```
bar = ax.bar(cities, revenue, color='lightyellow', edgecolor='white', width=0.3)
```

```
for bars in bar:  
    height = bars.get_height()  
    ax.text(bars.get_x() + bars.get_width()/2,  
            height + 50,  
            str(height),  
            ha='center',  
            va='bottom')
```

```
plt.show()
```



```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd
```

```
sns.set_theme(style="darkgrid")
df=sns.load_dataset('titanic')
print(df.head())
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

data = {
    "Athlete": ["Usain Bolt", "Michael Phelps", "Simone Biles", "Neeraj Chopra", "Allyson Felix",
               "Carl Lewis", "Katie Ledecky", "Abhinav Bindra", "Mo Farah", "Shelly-Ann Fraser"],
    "Country": ["Jamaica", "USA", "USA", "India", "USA",
               "USA", "USA", "India", "UK", "Jamaica"],
    "Sport": ["Athletics", "Swimming", "Gymnastics", "Athletics", "Athletics",
              "Athletics", "Swimming", "Shooting", "Athletics", "Athletics"],
    "Gold": [8, 23, 7, 1, 7, 9, 7, 1, 4, 3],
    "Silver": [0, 3, 2, 0, 3, 1, 3, 0, 0, 4],
    "Bronze": [0, 2, 2, 0, None, 0, 1, 0, 0, 0] # Missing value
}

df = pd.DataFrame(data)
print(df)
# Fill missing values
df["Bronze"] = df["Bronze"].fillna(0)

# Convert to integer
df["Bronze"] = df["Bronze"].astype(int)

print("\nCleaned Data:\n")
print(df)
medals = np.array(df[["Gold", "Silver", "Bronze"]])

# Total medals
df["Total"] = medals.sum(axis=1)

print("\nWith Total Medals:\n")
print(df)
medals = np.array(df[["Gold", "Silver", "Bronze"]])
```

```
# Total medals
df["Total"] = medals.sum(axis=1)

print("\nWith Total Medals:\n")
print(df)
top = df.sort_values(by="Total", ascending=False)
print("Top Athlete:\n", top.iloc[0])
country_total = df.groupby("Country")["Total"].sum()
print(country_total)
print(df["Sport"].value_counts())
print("Average Gold:", df["Gold"].mean())
india = df[df["Country"] == "India"]["Total"].sum()
print("India Total Medals:", india)
plt.figure(figsize=(6,4))
plt.hist(df["Total"], edgecolor='black')

plt.title("Distribution of Total Medals")
plt.xlabel("Total Medals")
plt.ylabel("Count")

plt.show()
country_total.plot(kind='bar')

plt.title("Total Medals by Country")
plt.xlabel("Country")
plt.ylabel("Medals")

plt.xticks(rotation=45)
plt.show()
df_sorted = df.sort_values(by="Total", ascending=False)

plt.figure(figsize=(8,5))

plt.plot(df_sorted["Athlete"], df_sorted["Total"], marker='o')

plt.title("Total Medals by Athlete (Line Chart)")
plt.xlabel("Athlete Name")
plt.ylabel("Total Medals")

plt.xticks(rotation=45)
plt.grid()

plt.show()
```


	Athlete	Country	Sport	Gold	Silver	Bronze
0	Usain Bolt	Jamaica	Athletics	8	0	0.0
1	Michael Phelps	USA	Swimming	23	3	2.0
2	Simone Biles	USA	Gymnastics	7	2	2.0
3	Neeraj Chopra	India	Athletics	1	0	0.0
4	Allyson Felix	USA	Athletics	7	3	NaN
5	Carl Lewis	USA	Athletics	9	1	0.0
6	Katie Ledecky	USA	Swimming	7	3	1.0
7	Abhinav Bindra	India	Shooting	1	0	0.0
8	Mo Farah	UK	Athletics	4	0	0.0
9	Shelly-Ann Fraser	Jamaica	Athletics	3	4	0.0

Cleaned Data:

	Athlete	Country	Sport	Gold	Silver	Bronze
0	Usain Bolt	Jamaica	Athletics	8	0	0
1	Michael Phelps	USA	Swimming	23	3	2
2	Simone Biles	USA	Gymnastics	7	2	2
3	Neeraj Chopra	India	Athletics	1	0	0
4	Allyson Felix	USA	Athletics	7	3	0
5	Carl Lewis	USA	Athletics	9	1	0
6	Katie Ledecky	USA	Swimming	7	3	1
7	Abhinav Bindra	India	Shooting	1	0	0
8	Mo Farah	UK	Athletics	4	0	0
9	Shelly-Ann Fraser	Jamaica	Athletics	3	4	0

With Total Medals:

	Athlete	Country	Sport	Gold	Silver	Bronze	Total
0	Usain Bolt	Jamaica	Athletics	8	0	0	8
1	Michael Phelps	USA	Swimming	23	3	2	28
2	Simone Biles	USA	Gymnastics	7	2	2	11
3	Neeraj Chopra	India	Athletics	1	0	0	1
4	Allyson Felix	USA	Athletics	7	3	0	10
5	Carl Lewis	USA	Athletics	9	1	0	10
6	Katie Ledecky	USA	Swimming	7	3	1	11
7	Abhinav Bindra	India	Shooting	1	0	0	1
8	Mo Farah	UK	Athletics	4	0	0	4
9	Shelly-Ann Fraser	Jamaica	Athletics	3	4	0	7

With Total Medals:

	Athlete	Country	Sport	Gold	Silver	Bronze	Total
0	Usain Bolt	Jamaica	Athletics	8	0	0	8
1	Michael Phelps	USA	Swimming	23	3	2	28
2	Simone Biles	USA	Gymnastics	7	2	2	11
3	Neeraj Chopra	India	Athletics	1	0	0	1
4	Allyson Felix	USA	Athletics	7	3	0	10
5	Carl Lewis	USA	Athletics	9	1	0	10
6	Katie Ledecky	USA	Swimming	7	3	1	11
7	Abhinav Bindra	India	Shooting	1	0	0	1
8	Mo Farah	UK	Athletics	4	0	0	4
9	Shelly-Ann Fraser	Jamaica	Athletics	3	4	0	7

Top Athlete:

```
Athlete    Michael Phelps
Country      USA
Sport        Swimming
Gold         23
Silver        3
Bronze        2
Total        28
```

Name: 1, dtype: object

Country

```
India      2
Jamaica    15
UK          4
USA        70
```

Name: Total, dtype: int64

Sport

```
# IMPORT LIBRARIES
```

```
import pandas as pd
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
sns.set_theme(style='whitegrid')
```

```
# LOAD DATASET
```

```
data = {
```

```
    "Athlete": ["Usain Bolt", "Michael Phelps", "Simone Biles", "Neeraj Chopra", "Allyson Felix",
               "Carl Lewis", "Katie Ledecky", "Abhinav Bindra", "Mo Farah", "Shelly-Ann Fraser"],
```

```
    "Country": ["Jamaica", "USA", "USA", "India", "USA",
               "USA", "USA", "India", "UK", "Jamaica"],
```

```
    "Sport": ["Athletics", "Swimming", "Gymnastics", "Athletics", "Athletics",
              "Athletics", "Swimming", "Shooting", "Athletics", "Athletics"],
```

```

"Gold": [8, 23, 7, 1, 7, 9, 7, 1, 4, 3],
"Silver": [0, 3, 2, 0, 3, 1, 3, 0, 0, 4],
"Bronze": [0, 2, 2, 0, None, 0, 1, 0, 0, 0]
}

df = pd.DataFrame(data)

print(df.shape)
print(df.dtypes)
print(df.head(3))

```

```

(10, 6)
Athlete      object
Country      object
Sport        object
Gold         int64
Silver       int64
Bronze       float64
dtype: object
   Athlete Country      Sport  Gold  Silver  Bronze
0  Usain Bolt  Jamaica  Athletics    8      0    0.0
1 Michael Phelps    USA    Swimming   23      3    2.0
2  Simone Biles    USA    Gymnastics    7      2    2.0

```

```

# STEP 1: IDENTIFY MISSING VALUES
print(df.isnull().sum())

# STEP 2: CLEAN DATA
df['Bronze'] = df['Bronze'].fillna(0)

# STEP 3: FIX DATA TYPES
df['Bronze'] = df['Bronze'].astype('int')

# STEP 4: VERIFY CLEANING
print('Missing after cleaning:', df.isnull().sum().sum())
print(df.describe())

# FEATURE ENGINEERING

```